INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM

PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree

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BONAFIDE CERTIFICATE

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DECLARATION

I affirm that the project work titled **"Intelligent Student Doubt Resolution System"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Biotechnology** is the record of original work done by me under the guidance of **Dr PAVITHRA M K S**, Associate Professor, Department of Biotechnology. It has not formed a part of any other project work(s) submitted for the award of any degree or diploma, either in this or any other University.

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ABSTRACT

The Intelligent Student Doubt Resolution System has been designed in response to the ever-increasing demand of an effective, technology-based academic assistance in the schools. Much of the conventional doubt resolution process is still based on manual communication, pre-established office timings, and email responses, which are time-consuming, haphazard and fail to offer students instant assistance when they require them the most. Students are frequently unable to receive the answer to their academic questions in a timely manner, whereas faculty members have a hard time dealing with numerous requests of students and responding to them effectively. The Intelligent Student Doubt Resolution System is the answer to these issues; it is a single system, which includes real-time chat, AI-based support, reserving slots, and sharing the study content. The site records key interactions between the students and staffs such as real-time chat sessions, slot bookings, AI chatbot conversational, and download of materials. These interactions are organized in a smooth user experience with specific dashboards of students and staff. The system has a secure authentication using JWT tokens and Google OAuth, role-based access control, real-time messaging using Socket.io, and AI-enhanced responses using Google Gemini AI. The intelligent chatbot allows students to immediately clarify concerns, and book a one-on-one meeting with staff. Employees have an opportunity to control their availability, post studying resources, and interact with students successfully. The site is developed with the MERN stack to provide a state of the art, scalable and responsive platform. React.js frontend consists of two applications, one student and one staff. The backend, which runs on Node.js and Express.js, provides authentication, real-time communication, integration of AI, file uploads, and API management. All in all, the Intelligent Student Doubt Resolution System is a reliable and user-friendly system that is intelligent enough to facilitate a better academic experience of students and faculty.

Keywords: Student Doubt Resolution, Real-Time Chat, MERN Stack, AI Chatbot, Google Gemini AI.

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CHAPTER 1

INTRODUCTION

Over recent years, the education sector has been embracing rapidly digital technologies with the aim of enhancing learning processes and academic support systems. Nevertheless, most schools continue to use manual, outdated, and traditional ways of addressing student inquiries, including physical office hours, email communication, and in-person meetings. The methods are usually slow, not available outside working hours and uneven in quality. Lack of access to prompt response to academic questions is a common problem among students that results in frustration, decreased learning effectiveness, and low academic outcomes. To overcome these difficulties, a smart and integrated platform that allows students to effectively resolve their doubts in a short period of time by using a combination of real-time communication and smart automation is highly demanded.

The Intelligent Student Doubt Resolution System was created to address these issues and provide an all-inclusive academic support system to students and staff. This platform aims to ensure a smooth communication between students and teaching staff via real-time chat, slot reservation, and AI-assisted doubt clearing. It documents all the important interactions such as chat sessions, slot booking, material uploads and AI response to enable a well-structured and orderly academic environment. The system assists institutions to offer superior and more efficient academic support to their students by integrating real-time messages and smart automation.

The site is coded on MERN stack that comprises of MongoDB, express.js, React.js and Node.js. It is made up of two distinct portals, a Student Portal, and a Staff Portal, which communicate via a common backend API and real-time Socket.io communication. Students will be able to explore the staff, reserve consultation time, chat with staff, communicate with an AI chatbot to immediately clarify doubts, and access study resources. The staff members are able to schedule their available times, answer student questions, post study materials by subject and topic, and track their chat feeds. Secure login with both JWT tokens and Google OAuth is also supported by the

system, which ensures both students and faculty members have a secure and safe user experience.

1.1 PURPOSE

The primary objective of this project is to develop an intelligent academic support platform that will merge a real-time interaction, the use of AI to resolve doubts, and systematic sharing of resources into one cohesive system. The platform will save time that students need to wait to have their academic queries answered besides lessening the manual labor of the other staff members who have to attend to various student queries at the same time.

In contrast to more traditional academic support systems, which use physical meetings and email communications, this platform provides students with immediate access to support using an AI chatbot based on Google Gemini AI, and allows them to schedule consultation sessions with staff and communicate in real-time. Employees can also have their own portal where they can make and manage their time slots, interact with students, post study materials, and track their engagements through a well-organized and user-friendly dashboard.

The system also facilitates organized sharing of materials where staff can post study materials which are classified according to subject and topic. Whenever they require it, students are able to easily browse, view and download these materials. This saves the hassle of sharing materials via email or handouts and makes learning resources available at all times. On the whole, the platform can contribute greatly to the quality, speed and effectiveness of academic support in educational facilities.

1.2 PROBLEM IDENTIFICATION

Most of the existing educational assistance programs do not have the capability of giving students instant and regular assistance with their educational inquiries. The majority of the institutions rely on predetermined office hours, emailing, and face to face meetings, which are restricted by time, availability, and geographical factors. Students are also made to wait hours or even days before their queries are responded to and this has a great effect on the way students learn. It does not have a centralized

system through which students can access staff, receive immediate responses using AI, make reservations, and access study materials all in one platform.

The other critical problem is absence of real time communication tools at the academic level. The old systems lack live messaging and real-time file sharing among students and employees. This complicates the ability of students to express their suspicions using supporting documents, diagrams, or images in the course of a chat session. Moreover, employees frequently cannot cope with serving a number of student requests simultaneously, which predetermines low quality of responses and lost inquiries. This issue is further aggravated by the lack of a formal slot booking system which the students do not have a clear means of booking a specific time to meet their teacher.

The Intelligent Student Doubt Resolution System solves all of these issues by offering a single platform, which will support real-time communication with the help of file attachments, doubt resolving based on AI with the help of the Google Gemini AI, book and manage slots, organize sharing of materials, and providing secure authentication. The system itself automatically handles user sessions, logs interactions and makes sure that the students and staff members have access to the correct and up-to-date information at all times. This minimizes the manual coordination, enhances the quality of response and makes academic support in the educational institution more efficient.

1.3 EXPECTED OUTCOME OF THE PROJECT

This project is expected to deliver a smart and efficient platform of academic support that will be useful to students and staff. The system will be designed to offer quicker and more dependable study assistance to students and lessen the manual coordination load on staff individuals.

- **Centralized Academic Support Platform:** The platform is a single system where students can chat in real-time, use the assistance of an AI chatbot, and book slots, as well as study materials, in the same place. Employees will have the opportunity to plan their slots, post materials, and interact with learners via a hub. This saves the complexity of having several independent systems and

enhances co-ordination. It also makes all the academic support activities to be organized and monitored within the same environment.

- **Real-Time Communication and AI-Powered Doubt Resolution:** The system will provide real-time communication via Socket.io, which will enable students and staff to communicate messages, images, and files in real-time. The Google Gemini AI-powered integrated AI chatbot offers immediate answers to frequently asked questions in academics, eliminating the process of students waiting to be responded to by a staff member. With this live chat and AI support, students will never be deprived of assistance, no matter how late the day is. The chatbot also provides formatted Markdown responses to make them easier to read.
- **Organized Slot Booking and Management:** Staff members are able to create and manage consultation slots, and students have access to available slots, and can create and reserve them easily. This removes the confusion of having informal schedule and both students and staff know what they are committing to in terms of consulting. Students will also be able to make cancellations and see their scheduled sessions on their dashboard. This organized strategy will enhance time management among students and teaching personnel.
- **Improved Student Learning Experience:** The site has a clean, responsive, and user-friendly interface to students. The availability of features like subject-based browsing of materials, real-time notifications, artificial intelligence chat, and slot booking make the academic support process straightforward and efficient. Students will be able to use all the platform features using any device, such as desktops, tablets, and mobile phones. An improved educational experience enhances student satisfaction, learning outcomes, and promotes regular use of the platform.
- **Scalability and Future Expansion:** The project is developed in a scalable and modular manner, and we can readily add new features onto it in the future. Other features like integration with video conferencing, peer-to-peer learning forums, assessment tools, notification systems, and analytics dashboards can be added

subsequently without significant alterations to the current system. The scalable design will enable the platform to expand as the needs of the institution increase and accommodate more students and staff members in the future.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW OF EXISTING METHODS

Kumar et al., (2022) examined AI-based chatbot systems to resolve doubt in e-learning platforms. Their paper emphasized the use of natural language processing and machine learning to comprehend student queries and give real-time responses. The authors presented a conclusion that AI chatbots can drastically decrease the amount of time to respond to typical academic doubts and facilitate self-paced learning, which is consistent with the AI-based doubt resolution of the proposed system.

Thomas and George (2023) investigated the use of Socket.io as a real-time communication system between students and the faculty. Their study highlighted how event-based communication can be used to minimize delays and enhance the responsiveness of educational chat systems. In the study, other characteristics were also brought to the fore like unread message notification and file sharing support that are the direct support of the real time chat component in the proposed platform.

Singh and Patel (2022) researched the topic of JWT-based secure authentication and role-based access controls in educational web applications. Their contribution showed the way JSON Web Tokens can be used to offer safe and stateless authentication in addition to best isolating student and staff access privileges. This paper justifies the secure login and access control facilities that would be part of the proposed system.

Nair and Krishnamurthy (2023) examined Google OAuth as a part of academic web applications to provide secure and convenient external authentication. Their results revealed that OAuth-based authentication enhances security, decreases complexity of registration, and improves user ease. This is directly connected with the Google sign-in feature of the proposed system to be able to easily access it.

Johnson and Williams (2023) talked about the academic consultation slot booking and appointment management systems. Their study emphasized the role of online booking of appointments in alleviating the conflicts and enhancing access to

faculty services among students. This will be in line with the slot booking option proposed in the proposed system to facilitate the staff-student consultation.

Ramesh and Iyer (2022) compared MongoDB and Mongoose in scalable data management in educational applications. They highlighted how NoSQL databases are flexible and efficient in managing various educational data including chats, bookings, and learning materials. This facilitates database design and scalability objectives of the proposed system.

George and Kumar (2023) introduced a study-on-study material management and distribution systems in online academic settings. According to their results, organized content posting and categorized access enhance the effectiveness of learning and availability of resources. This justifies the study material module in the proposed platform.

Lee and Park (2024) studied the principle of responsive UI design of student-centered academic portals in React.js and Bootstrap. Their study revealed that easy-to-navigate designs, quick loading, and mobile interfaces were essential in enhancing user experiences. This is in line with the frontend design strategy applied in the proposed system.

Balachandran and Suresh (2024) examined the implementation of Google Generative AI within chat systems in education to provide real-time support in academic activities. Their research has shown the potential of generative AI to comprehend context and give correct answers and save faculty time. This is a great affirmation of the AI integration of the proposed student doubt resolution system.

Verma and Rao (2023) investigated the application of Node.js and Express.js to create scalable backend systems in education. Their article highlighted the advantage of asynchronous processing, RESTful APIs, and architecting based on the middleware to ensure performance and reliability. This justifies the backend development framework that will be adopted in the proposed system.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

The INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM platform is designed to provide a smart and contemporary academic support platform that unites the students with the personnel via real-time communication, AI-based doubt resolution, slot scheduling, and structured sharing of study material. It will not only improve the student learning process but will also assist staff to handle their academic support roles more effectively. This system is highly interactive and can be used to offer intelligent support to students in various ways unlike in a typical academic platform which is primarily used to display information and give students access to non-interactive resources.

- The platform captures all important interactions between students and staff, including:
  - Real-time chat messages
  - Slot bookings
  - AI chatbot conversations
  - Study material downloads
- The system helps educational institutions to:
  - Improve response quality
  - Reduce administrative workload
  - Enhance overall academic support

3.1 OBJECTIVES OF PROPOSED WORK

The overall objective of the INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM platform is to create an automated and real time academic support system which facilitates easy doubt solving among the students and provides staff members with access to efficient communication and resource management

software. The project aims at enhancing the quality of academic interaction, lessening the manual coordination, and assisting students access help whenever they desire.

The following are the precise objectives of the study

- **Live Doubt Resolution System via Chat and AI:** The site offers students two options to resolve their doubts in academics: live chat with employees and an AI chatbot that is an instance of the Google Gemini AI. In chat sessions, students are able to exchange text messages, photos and file attachments to facilitate rich and detailed academic discussion. The AI chatbot answers frequent academic questions in real-time, 24/7, and in a clear form with Markdown rendering. This two-sided strategy will provide students with a means to seek assistance at any time, regardless of the immediate availability of a staff member.
- **Structured Slot Booking and Consultation Management:** The system will enable the staff members to create and manage the time slots of consultation, whereas students will be able to view the available slots and book a session that would fit their schedule. Students are able to see their future appointments, cancel appointments in case of necessity and get the confirmation of their planned consultations. The staff will be able to update slot information, erase expired slots, and track the number of students who have made a booking of their sessions. This is an organized booking system that does away with all informal booking and also provides both students and staff with a clear and structured consultations schedule.
- **Shared Materials of studying:** The site allows members of staff to post study materials according to subjects and topics and students can easily access and download materials that are of interest to them. Staff is also able to update or delete materials where necessary and see a list of all the resources that have been uploaded previously. Students are able to preview and go through all the available materials, supported file types and also download them to study offline. This will remove the necessity of sharing materials via email and will make sure that learning materials are always in place and available via a centralized system.

- **Student and Staff Interactive Dashboards:** Both students and staff will be given dedicated student and staff dashboards that allow them to use features specific to their role and easily access all key platform capabilities. The student dashboard shows the profile of the student, features, including chat, slot booking, and materials available, and a brief review of what the student has done. Staff dashboard will display the future bookings, chat history, and material handling. Both dashboards will be user-friendly and easy to navigate, hence a great user experience to all users.
- **Secure Authentication and Role-Based Access:** The platform is compatible with both standard email and password authentication with JWT tokens and Google OAuth integration. Role-based access control will ensure that students and staff will only view and interact with features pertinent to their role. The unauthorized access to the protected pages is prevented by the usage of the private route, and all the passwords are stored in the secure way with the help of the bcrypt hash. This will make sure that any data of user and academic interactions is secure and confidential over the platform.
- **Scalability and Future Growth:** The platform is implemented in a modular manner that allows adding more features in the future. Video conferencing, peer-to-peer discussion forums, automated quiz systems, performance analytics, and push notifications are features which can be added later on without the existing structure. The system architecture is scalable and flexible to accommodate additional users, materials and interactions as the institution expands.

3.2 SYSTEM ARCHITECTURE

The INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM platform system architecture is broken down into three key layers, namely, Frontend Layer, Backend Layer, and Database Layer. These layers are integrated to provide a smooth communication process, fast processing and real time updates.

3.2.1 System Architecture Overview

- **User Interaction Layer (Frontend):** The frontend is created with React.js and offers student and staff interfaces. The Student Portal allows students to browse the staff available, book slots, chat real-time, visit the AI chatbot, and download study materials. Through the Staff Portal, staff members manage their consultation times, reply to student chat messages, upload study materials and see their dashboard. Both portals are based on React components to create reusable and responsive interfaces, styled and laid out with Bootstrap 5 and custom CSS.
- **Application Logic Layer (Backend):** The backend is created with the help of Node.js and Express.js. It does authentication, API requests, real-time Socket.io, file uploads using Multer, AI chatbot answers using Google Generative AI and slot and material management. The backend includes authentication, chat, slots, materials and doubt routes and controllers. It handles all client requests and provides the necessary information via secure RESTful APIs secured with the help of JWT middleware.
- **Database Layer (MongoDB):** MongoDB will be deployed as a database in which the profiles of users, chat messages, bookings of slots, study materials, and records of doubts will be stored. Users, chats, slots, and materials are separated into separate collections. MongoDB has a flexible document-based design that allows the storage and retrieval of all data on the platform and, in the chat logs (complex nested objects), file attachment metadata, etc.
- **Real-Time Communication Layer:** Socket.io will be used to facilitate two-way real-time communication between staff and students. When a student sends a message, the server picks up the message and stores it to MongoDB and sends it to the socket connection of the recipient in real time. The number of unread messages is monitored and updated dynamically with socket events and polling to make sure that the users never miss new messages.
- **AI Integration Layer:** The platform uses the Google Generative AI SDK to use Google Gemini AI to drive the student-facing AI chatbot. When a student enters

a query is the chatbot, the backend forwards the query to the Gemini API and provides the answer to the student chat interface. The answers are made in Markdown formatting in order to enhance readability and clarity.

3.3 FLOW DIAGRAM

The Flow Diagram is shown in figure 3.1

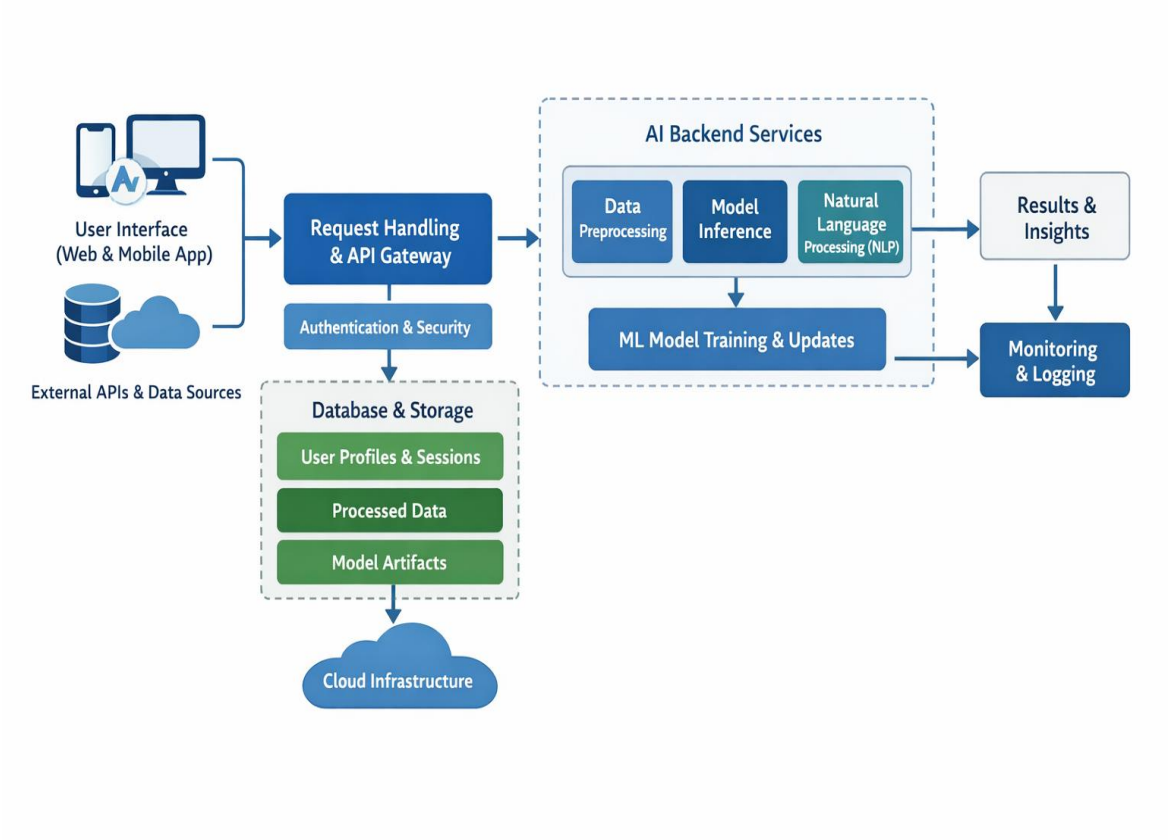

Figure 3.1 Flow Diagram

3.4 METHODOLOGY

The Intelligent Student Doubt Resolution System is designed in a systematic and guided approach that makes it to be efficiently designed, developed and deployed. The methodology starts with a requirement analysis in which there is a thorough examination of the needs of the users to find out their challenges in academic doubt-solving. According to (Kumar et al., 2022), which includes AI-based chatbot systems, has a great impact on the rapidity and precision of the resolution of student queries, which inspired the inclusion of AI capabilities in this project.

The system design is aimed at developing user friendly and responsive interfaces to both students and staff. Modern frameworks are used to develop the frontend to make it accessible and easy to use. It is as noted in (Lee, D., and Park, H. 2024), responsive UI design is an important factor in enhancing user interactions and experience in an academic platform. This principle was used in the formulation of intuitive dashboards and navigation structures within the system.

3.4.1 PLANNING

- **Find User Needs:** The first process was to get knowledge about the issues of students and staffs within the traditional academic support systems. Students usually have to go through extended periods of waiting in order to resolve doubts, have fewer access to faculty members not in their offices, and there is no centralized place to access study materials. Equally, faculty members find it difficult to handle various student inquiries, arrange consultations and allocating learning resources effectively. These results facilitated the establishment of the fundamental goals of the project (Balachandran and Suresh, 2024; George and Kumar, 2023; Johnson and Williams, 2023).
- **Project Objectives:** In line with the problems identified, the foremost goal was to create a smart academic support platform that includes real-time communication, AI-based doubt resolution, slot booking, and sharing of study material. The platform allows students to communicate with faculty, receive the help of an AI chatbot, reserve consultation, and access materials. To the staff, it offers resources to make bookings, contact students, and upload documents (Kumar et al., 2022; Thomas and George, 2023).
- **Designing the Platform:** The system was configured as two distinct portals, one being a Student Portal and the other a Staff Portal that share a common backend and a database. The layout provides role-specific functionality and intuitive interface. Learners can use such features as chat, slots, materials, and AI chatbot and staff can use slots, upload content, and answer student questions (Lee and Park, 2024; Nguyen and Chen, 2024).

- **Adding Core Functionalities:** The system has incorporated several core features, such as JWT-based authentication, Google OAuth login, real-time chat with Socket.io and file sharing, AI chatbot integration with Google Gemini, slot booking and management, and upload/download of study material. MongoDB was used to store and manage all interactions efficiently (Singh & Patel, 2022; Nair & Krishnamurthy, 2023; Patel & Kumar, 2023; Ramesh & Iyer, 2022).
- **Project Timeline:** An orderly schedule was drawn up to facilitate orderly progress. The stages of the project were requirement analysis, UI design, backend development, database setup, Socket.io integration, AI integration, testing, and deployment. The phases were done and verified before transitioning to the next (Verma & Rao, 2023).
- **Testing and Feedback:** Some of the testing strategies which were included in the project are unit testing, integration testing, security testing and usability testing. The feedback by the users was deemed to enhance the usability of the system and the overall experience (Sharma and Nair, 2025).

3.4.2 DESIGNING PHASE

Design stage concentrated on designing attractive, user friendly and responsive interfaces that would be attractive to both students and staff. This was aimed at making it easy to use and to be available on devices.

- **Developing a Mockup** The entire prototype of the platform was created to visualize the user flow. It featured student login, dashboard, slot booking, chat, AI chatbot, materials section, staff login, staff dashboard, slot management and material upload. This assisted in detecting features that were missing and design flaws at an earlier stage of the development (Lee and Park, 2024).
- **Design Tools and Approach:** The interface was designed with modern UI components including cards, tables and menus. The student dashboard allows the student to access all significant features quickly, whereas the staff dashboard shows booking summaries and the latest interactions. Responsive layouts were made with Bootstrap 5, and it was supplemented with custom CSS to make further styling and improvements (Lee and Park, 2024; Nguyen and Chen, 2024).

Customization of the Design and Visuals: The site has the same color schemes and graphics to ensure simplicity and clarity. Indications of messages and booking statuses are displayed as icons and badges. The design is compatible across mobile devices, tablets, and desktops, making it more accessible and user-friendly (Lee and Park, 2024).

3.4.3 TOOLS AND TECHNOLOGIES USED

Front-End Development

The frontend of the platform is built with React.js as it enables the platform to be created with reusable elements and dynamic user interfaces. It assists in providing a seamless and interactive user experience by loading only what needs to be updated to the screen without the load of the whole screen. It uses Bootstrap 5, HTML, CSS, and JavaScript as well as React.js to enhance the functionality and design of the platform. Student and staff portals are developed as two independent applications with React.js to allow both applications to run independently, sharing the same backend. State management and authentication state as well as theme preferences are handled with React Context API throughout the application.

Back-End Development

The coding of the backend is created with Node.js and Express.js. The node.js is fast and efficient in server-side processing and the express.js is easy to use in the API development and routing. They are together in handling all significant operations like user authentication, slot handling, material handling, AI chatbot, and real-time Socket.io communication. Multer is also used in the backend to process file uploads so that students and staff can share images and documents during chat sessions and staff can upload study material. Bcryptjs is used to hash passwords and JWT is used to authenticate with the use of a token. The Google Auth Library has been integrated with Google OAuth to provide an easy way to log in.

Real-Time Communication

The real-time bidirectional communication between the staff and students is provided through Socket.io. In case a user sends a message, the socket event is received by the server, the message is stored in MongoDB and emitted to the receiver. To keep

the number of unread messages updated, both socket events and polling are used to maintain the correct notification update. This gives it a smooth and instant messaging experience like the current messaging applications.

AI Integration

The platform incorporates Google Gemini AI with the Google Generative AI SDK of Node.js. When a student enters a query to the AI chatbot, the backend forwards the query to the Gemini API and displays the answer on the chat interface of the student. Answers are rendered in Markdown on the front-end, making them well-structured and clear-cut answers to academic questions.

Database Management

The database is MongoDB since it is flexible, scalable, and can accommodate various types of information in the form of documents in a JSON format. It holds user profiles, chat messages along with file attachment metadata, slot booking records and information on study material. Mongoose is adopted as the ODM to specify data structure and carry out database actions effectively.

3.4.4 TESTING AND QUALITY ASSURANCE

Testing is a significant aspect of the project as it will be used to verify that all features are accurate and as per the expectation of the users. Before the deployment, proper testing enhances reliability and security of the system.

- **Unit Testing:** Unit tests were done to check the functionality of each module of the platform individually. Registering, logging in, creating a slot, booking a slot, chatting, AI chatbot responding, and uploading materials were individually tested. As an illustration, the authentication module was tried using valid and invalid credentials whereas the chat module was tested to send and receive real-time messages.
- **Integration Testing:** The frontend, backend and database were carefully tested. APIs, real time socket.io events, slot management and AI chatbot responses were tested to provide smooth communication between the various system

components. Integration testing ensured that all the processes connected with each other did not have delays or loss of data.

- **Functionality Testing:** Various operations by the users were tested in various conditions to verify proper platform behavior. This involved registering students, Google OAuth staff login, slot booking and cancellation, real-time chat with file attachments, chatbot interaction with AI, and uploading and downloading material. The functionality testing ensured that everything was functional in both normal and edge-case scenarios.
- **Security Testing:** The site was also tested in the area of secure authentication, safe routes, and security of user information. The use of password hashing using bcrypt, validation of JWT tokens, and role-based access control as well as integration with Google OAuth were confirmed to ensure unauthorized access. The system was also tested so that the students cannot access the staff functionality and those who are unauthenticated are redirected to a login page.
- **Performance Testing:** Multiple concurrent users were used to test the platform to determine stability when under load. The speed of loading pages, API response time, delivery of real-time messages, and file upload performance were measured and optimized. Stress testing identified that the system was able to support simultaneous connections without compromises in performance.
- **Usability Testing:** To make the platform easy and simple to use, usability testing was carried out to make the site easy to navigate by students and staff. The aspects of navigation, design, responsiveness, and feature discoverability were rated. Layout, button placement, color contrast, and mobile responsiveness were improved based on the feedback.

3.4.5 DEPLOYMENT

Deployment entailed transfer of the frontend portals, backend API and database to a live environment. Student Portal and Staff Portal were deployed on Netlify and redirects were set up to be compatible with React Router. The API was implemented on a cloud hosting platform like Render or Heroku and environment variables were set to MongoDB URI, JWT secret, Google API key, and CORS origins.

Prior to final deployment, end-to-end testing was done to verify that all features operated properly in the production environment. The platform was secured with the help of SSL certificates and secure database connections. The platform will be able to accommodate future modifications and upgrades with minimum downtime.

3.5 PROJECT OUTPUT

The project deliverable offers the students and the staff a full-scale and easy to use academic support platform. Students are allowed to explore the staff available, reserve a consultation session, chat in real-time with the ability to share files, communicate with the AI chatbot, and access study materials. Staffing members will be able to make and maintain slots, chat with students in real time, and post structured study materials. Everything is automatically registered and updated throughout the platform.

The student login page of the Intelligent Student Doubt Resolution System is given in Figure 3.2. This is the landing page of all registered students using the platform. It will include input boxes to the email address and password of the registered student, and a safe log-in button. Lack of an account will direct students to the registration page through the link shown below the form. There is also a Google Sign-In button, which enables students to get quickly logged in with their existing Google account. The page is completely responsive and functions well both on mobile and desktop.

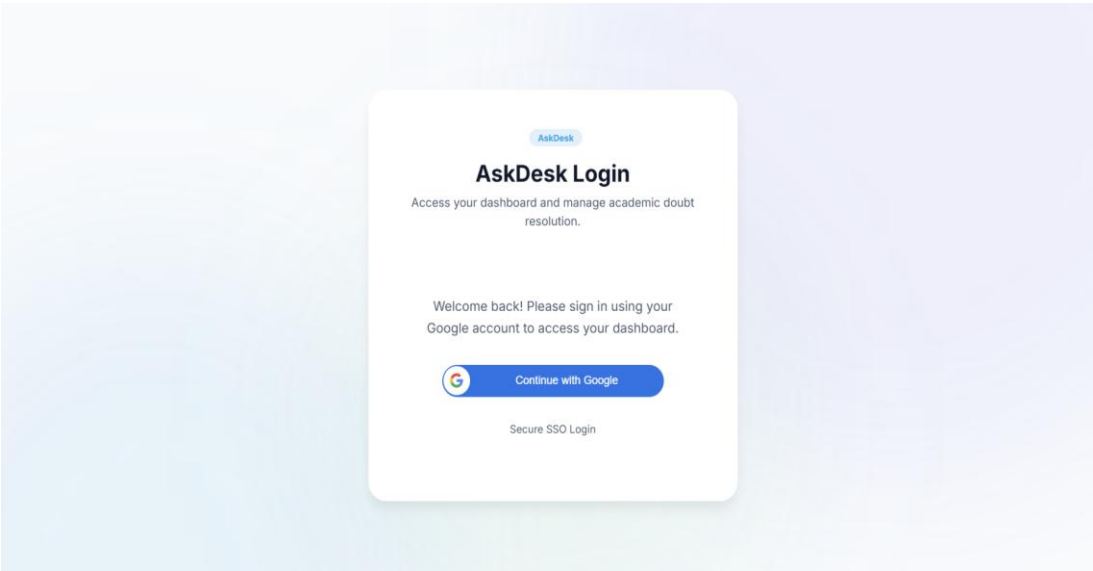

Figure 3.2 - Student Login

Figure 3.3 shows the staff login page of the Intelligent Student Doubt Resolution System. This page will be the secure gateway to all registered staffs that handle student suspicions, consultation schedules, and course materials. The login form contains input boxes which require the email address and password used by the staff member to log in, and the login button which verifies credentials and redirects to staff dashboard in case of successful verification. It is also conveniently accessible with a Google Sign-In option. The role-based authentication allows staff credentials to be authenticated independently of student accounts and stops staff-only features being accessed by staff.

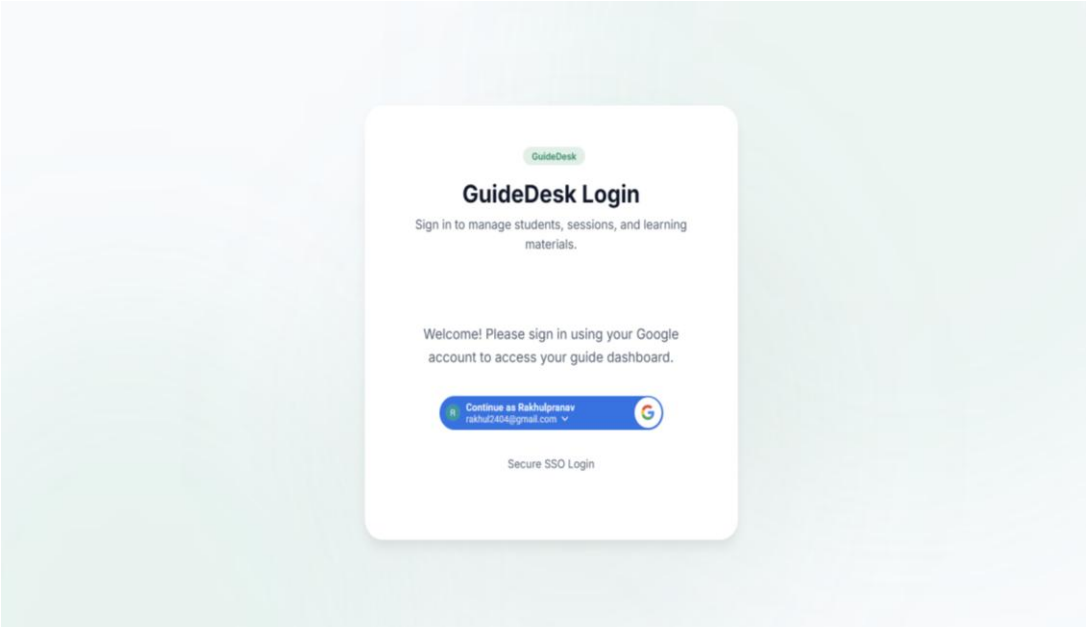

Figure 3.3 - Staff Login

CHAPTER 4

RESULT AND DISCUSSION

The Intelligent Student Doubt Resolution System was well implemented and all the modules scheduled worked well. The well-organized and modular design of software enhances the system reliability and maintainability as it is evident in the seamless nature of real-time communication, AI-based doubt-busting, slot booking, and managing of study materials integrated into the platform.

The system showed to perform efficiently upon testing, with various student and staff accounts to test all the features. Students could register and log in through standard authentication as well as Google OAuth, book slots, chat in real-time and share files, get AI responses and access study materials without issues. This is in line with the findings by Thomas and George (2023), who emphasized that real-time communication systems play a crucial role in increasing responsiveness and interaction with users.

The backend system also provided uniformity in the management and synchronization of data in all modules. The automatic updates available in the staff portal on student activities (messaging and slot booking) confirm the power of event-driven architecture as explained by Fielding and Taylor (2020) in the design of a system based on the REST.

All in all, the system fulfilled its goal of offering a reliable, scalable, and intelligent platform of academic support, which proves that the combination of latest web technologies and AI can greatly enhance the efficiency and access to educational support systems.

4.1 RESULTS

STUDENT PORTAL-HOME PAGE

The Student Home Page is shown in Figure 4.1. The main interface that is presented upon login. It allows having a centralized dashboard through which the students will be able to find their profile and go to the essential functionalities like chat, slot booking, AI chatbot, and study materials. A dashboard is well organized, which enhances ease of use and minimizes the complexity of navigation, which is an important concept in Human-Computer Interaction (Preece et al., 2015).

The overview section displays platform activities, allowing students to get a quick grasp of what can be done and what is being updated. This centralized access maximizes the user efficiency and interaction in digital learning environments (Nielsen, 2012).

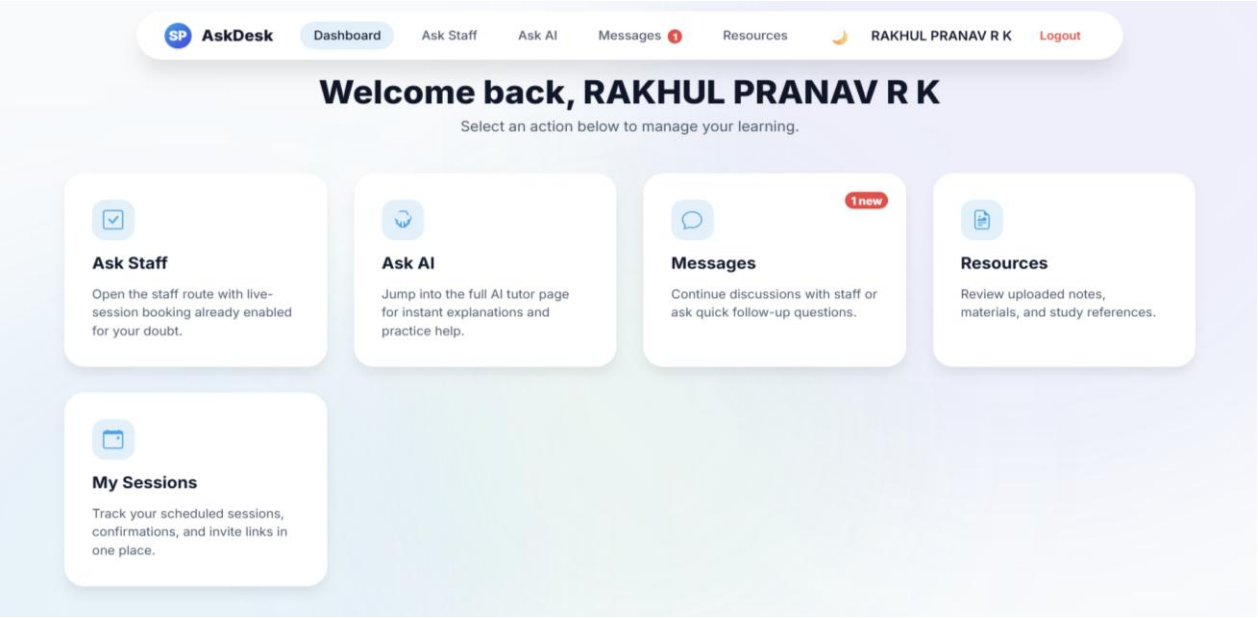

Figure 4.1: Student -Home Page

STUDENT PORTAL-SLOT BOOKING PAGE

The student-slot Booking page is shown in Figure 4.2. It allows the students to make appointments with members of staff. It guarantees real time visibility of available slots and real time updates at time of booking. Academic settings can enhance coordination and minimize conflicts with the help of real-time scheduling systems (Pressman and Maxim, 2020).

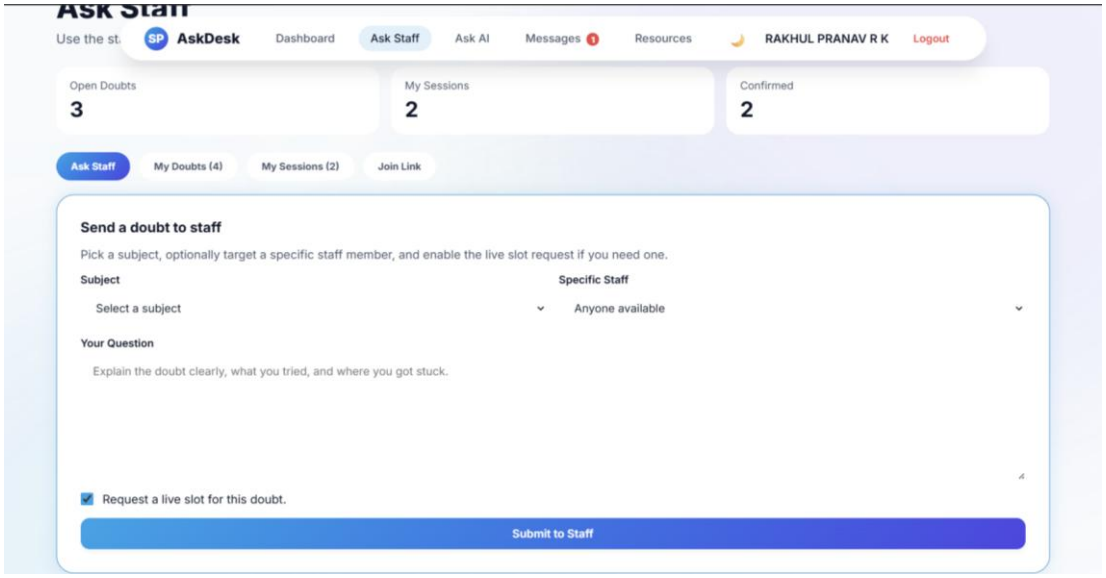

Figure 4.2: Student-Slot Booking Page

STUDENT PORTAL-AI CHATBOT INTERFACE

The student-Ai Chatbot is shown in figure 4.3. An intelligent chatbot system within the platform. The chatbot can answer academic questions instantly and in an orderly manner and is powered by Google Gemini. The chatbot allows Markdown formatting, which allows the straightforward display of such complex information as code fragments, lists, and headings. AI-based chatbots are gaining a growing popularity in education, where they are able to offer scalable and personalized assistance and enhance student engagement.

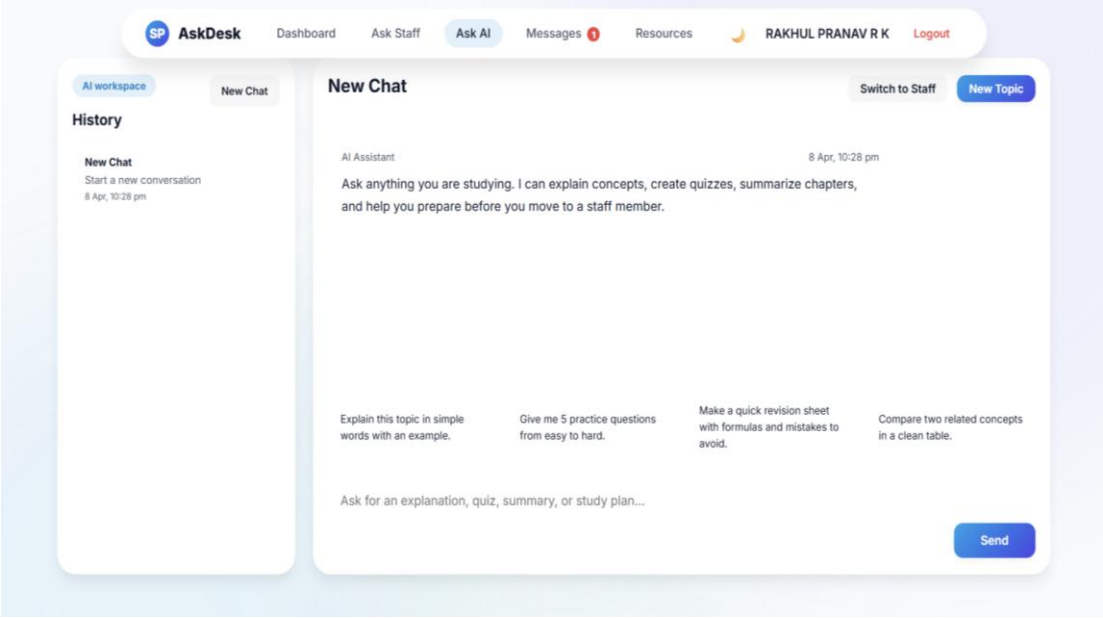

Figure 4.3: Student-Ai Chatbot Interface

STUDENT PORTAL-STUDY MATERIALS PAGE

The student study materials page is shown in figure 4.4 Resources are classified according to subject and topic, and also have metadata like title, upload date and description. These organized content management systems enhance the level of information retrieval and facilitate self-directed learning (Siemens, 2013). Materials can be easily downloaded by students thus allowing them easy and flexible access to academic materials.

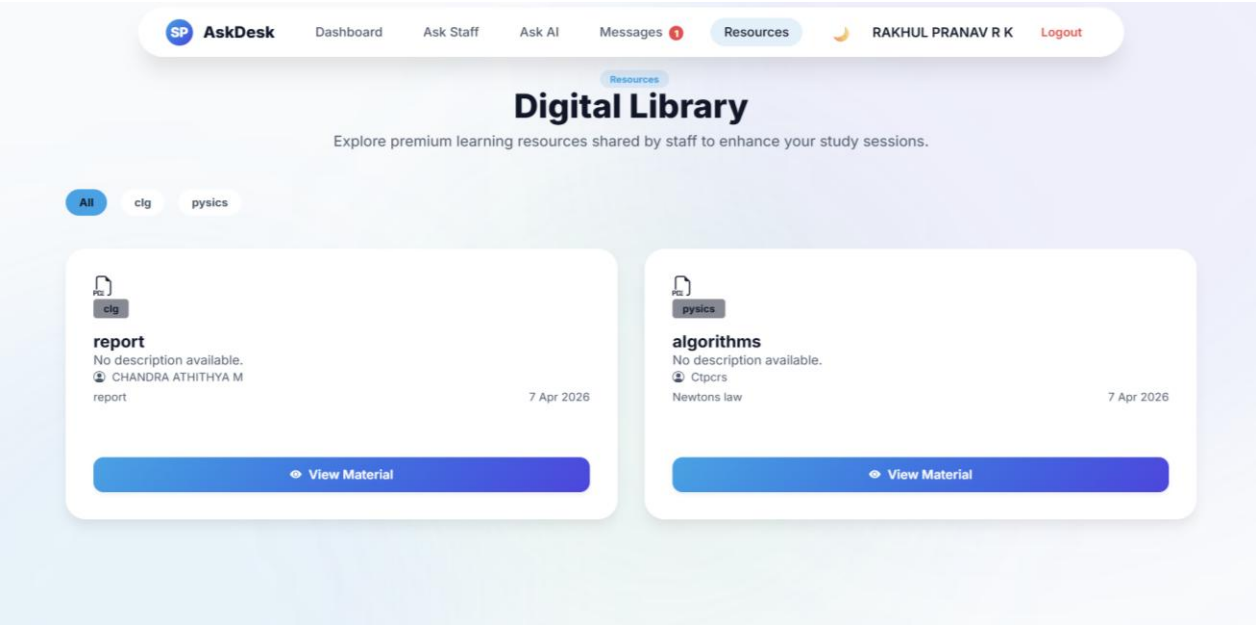

Figure 4.4: Student-Study Materials Page

STUDENT PORTAL-MESSAGES PAGE

The messages page of the student portal in shown in Figure 4.5. It helps in real time communication between the staff and students. It enables students to send and get messages, see the history of conversations and share files like documents or pictures. The system also provides real-time communication technologies to deliver instant messages and enhance speed of interaction and collaboration.IO). The organized chat interface improves clarity and enables students to follow discussions effectively, which leads to the effective provision of scholarly support and answer to uncertainties.

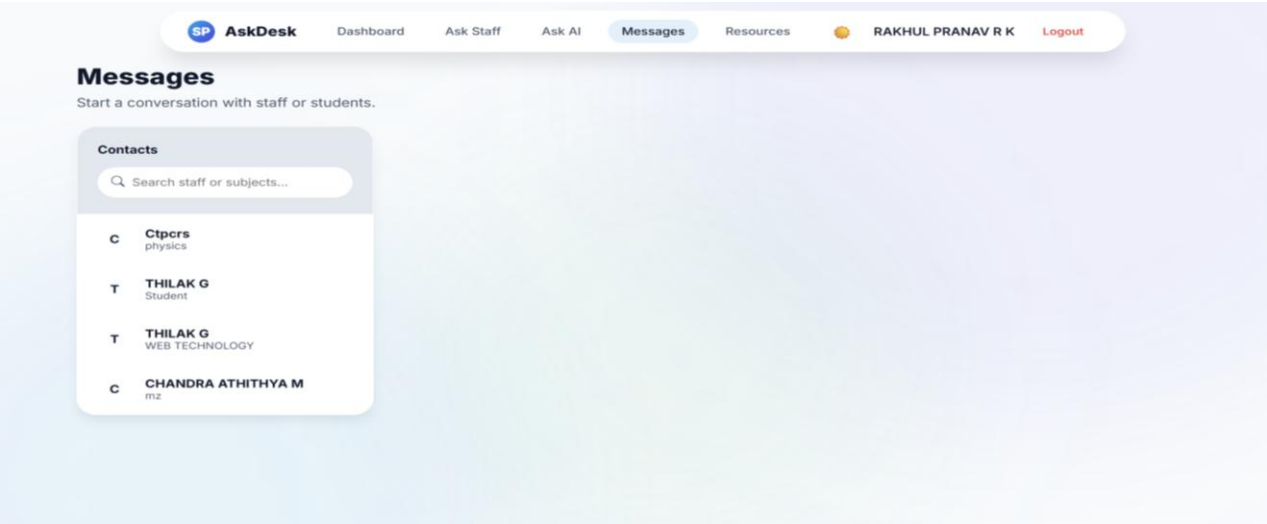

Figure 4.5: Student-Messages Page

STAFF PORTAL-DASHBOARD

The Staff-Dashboard is shown in figure 4.6. gives an overall view of the system activity such as future appointments, latest student messages, booked sessions and uploaded materials. Dashboards are critical in simplifying high amounts of data into consumable insights, enhancing decision-making and task management (Few, 2013). The interface enables navigating to essential functions like chat, slot management, and uploading resources quickly. The portal will allow the staff to manage doubts.

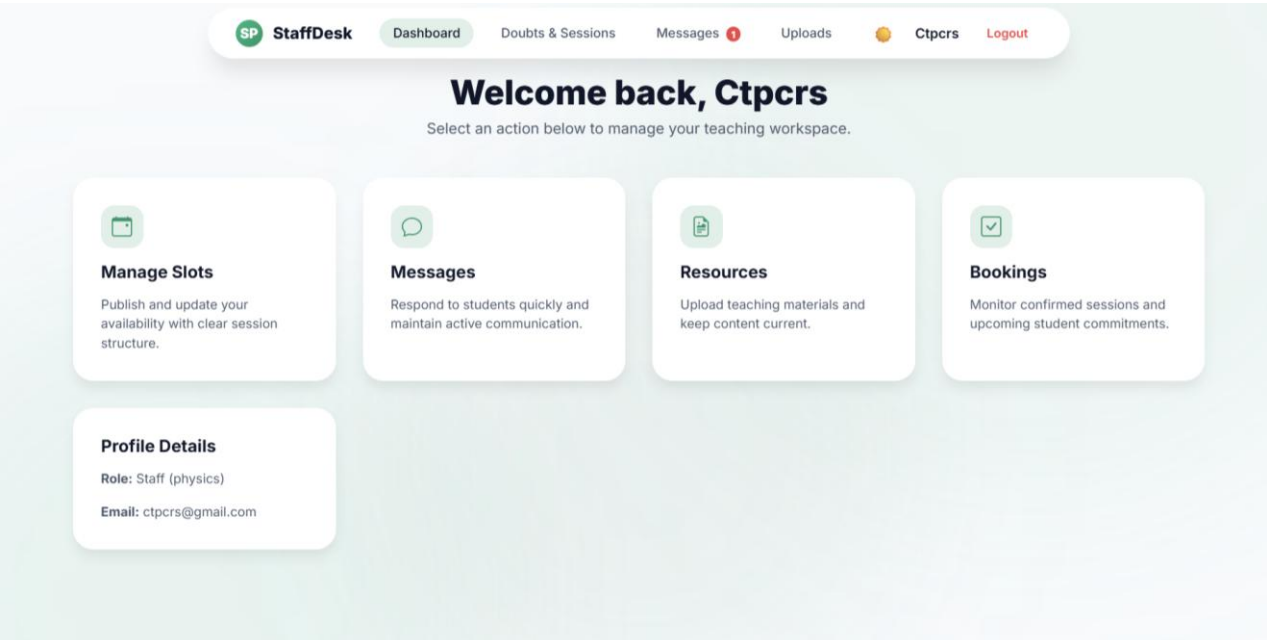

Figure 4.6: Staff-Dashboard

STAFF PORTAL-DOUBT MANAGEMENT

The Staff-Doubt Management is shown in figure 4.7. allows employees to effectively deal with student queries. Employees will be able to see the doubts that are getting in, respond immediately, and monitor past interactions. Such organized management of queries enhances the quality of response, as well as a systematic provision of academic support. The ability to share files also contributes to improved communication, particularly in complicated discussions (Pressman and Maxim, 2020).

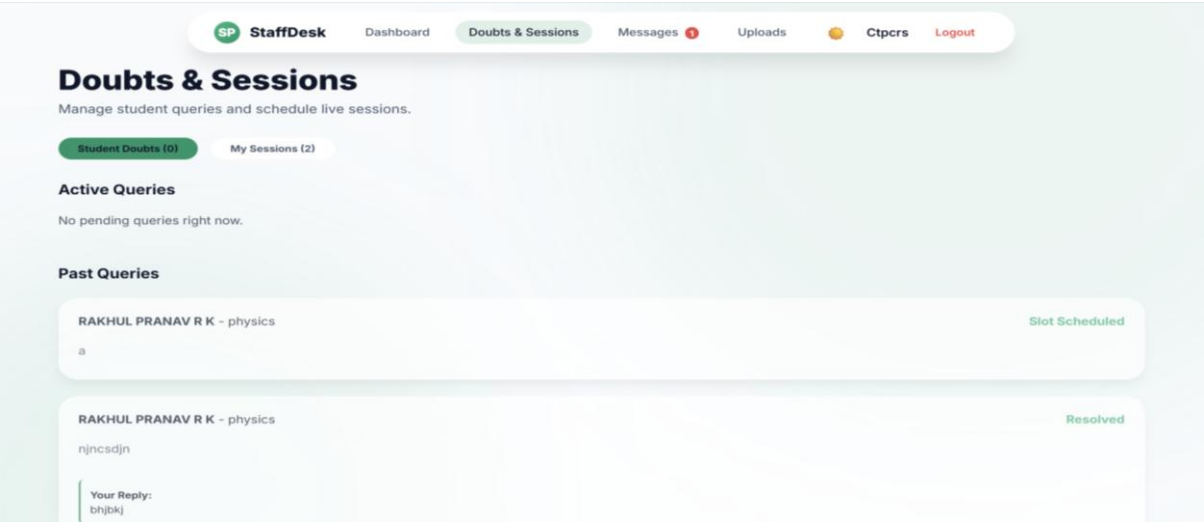

Figure 4.7: Staff-Doubt Management

STAFF PORTAL-CHAT PAGE

The Staff-Chat is shown in Figure 4.8. will enable one-on-one interactions with students via a chat program that is available in real-time. Employees are able to reply immediately, consult conversation history and exchange files as required. Educational systems are extremely responsive and collaborative based on the use of real-time communication platforms (Stallings, 2018). The structured chat design will provide effective monitoring and control of various student interactions.

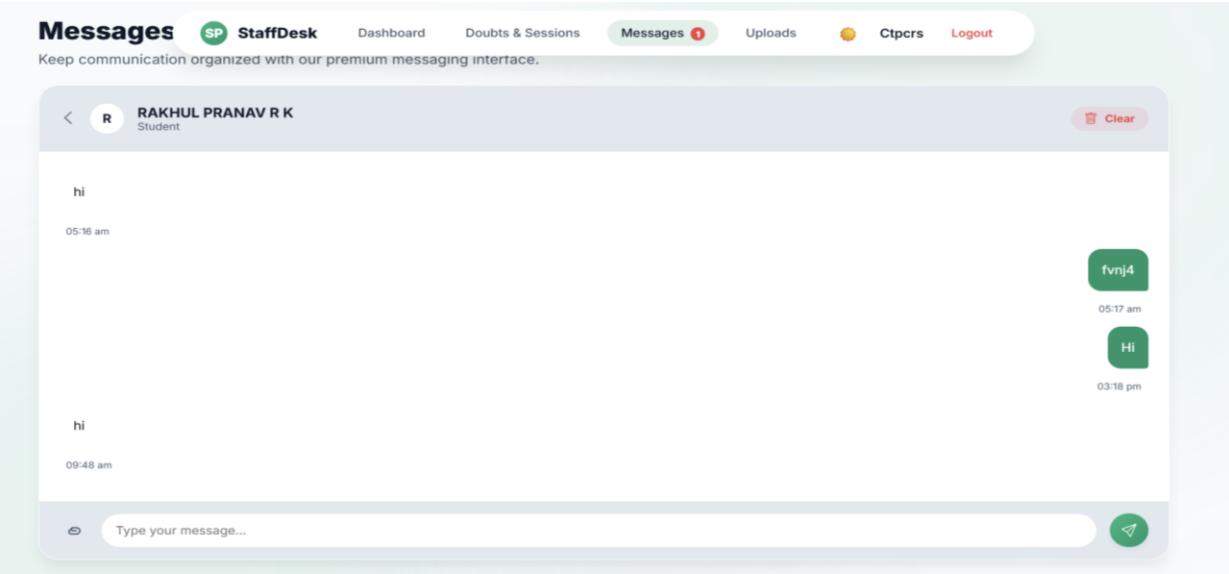

Figure 4.8: Staff-Chat

STAFF PORTAL-MATERIAL MANAGEMENT

The Staff- Chat and Material Management is shown in Figure 4.9. allows employees to add, update, and organize academic resources. Employees are able to access the materials uploaded before, modify information, and delete obsolete information. Also, the built-in chat system shows the list of student messages with read/unread tags, enhancing the process of communication monitoring. The content management systems are very important in the organization and maintenance of the educational resources in an effective manner to make them available and relevant (Siemens, 2013).

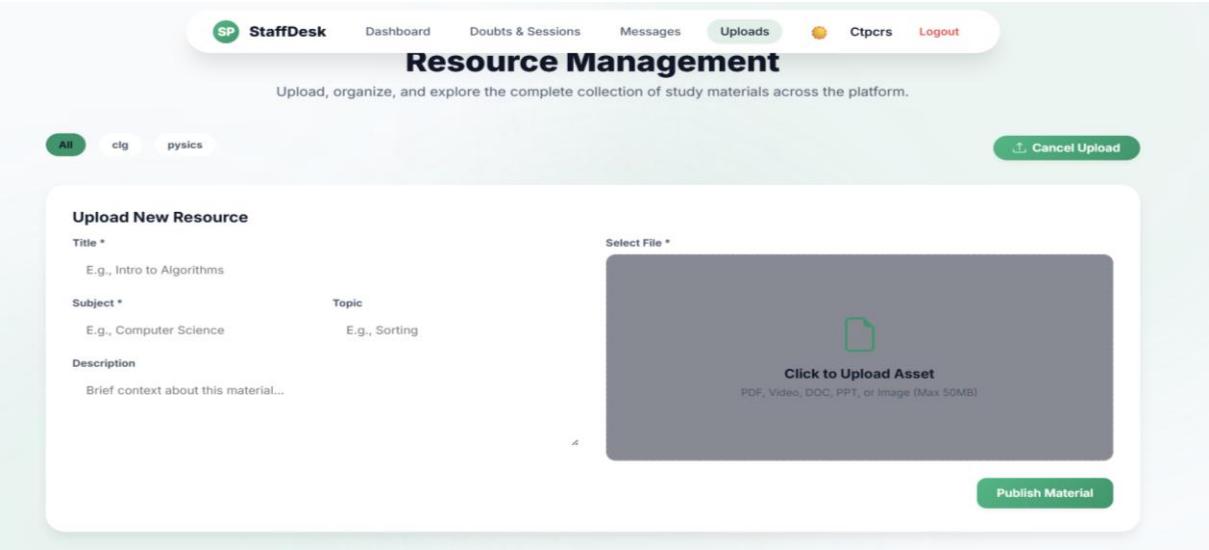

Figure 4.9: Staff- Chat and Material Management

4.1.1 FUNCTIONALITY REVIEW

Intelligent Student Doubt Resolution System is a system that gives the latest information to students and staff and allows real-time capabilities of the system. All the major student activities, like sending instant messages, making appointments, asking the AI chatbot questions, or downloading materials are immediately processed and reflected throughout the platform. Real-time systems enhance interactivity and real-time in learning platforms (Real-Time Systems) (Stallings, 2018).

Once a student messages, he/she is sent to the Socket.io server, stored in MongoDB and sent directly to the active connection of the recipient. Messages that have not been read are updated in real time and so the staff are never in the dark of new messages. The AI chatbot replies to queries posed by students using the Google Gemini API, responding to queries with structured Markdown responses in a few seconds. Accessibility and responsiveness in contemporary education are highly promoted through such AI-assisted support systems (Luckin et al., 2016).

Any changes made to the appointment schedule are immediately reflective in the student interface as well as the staff interface. The staff dashboard will allow the user to control the time of the appointment slots, keep a record of the interaction with students, plan learning materials and answer questions in real-time. This is because continuous data synchronization guarantees a reliable system even when using it at high user loads. Also, all discussions and appointments, as well as uploaded documents are safely stored, with timestamps to be used later, to facilitate traceability and accountability (Pressman and Maxim, 2020).

4.1.2 USER INTERFACE AND EXPERIENCE

The platform is user-friendly and will be developed with the usability in mind and a smooth user experience, so that students and staff can effectively use the system. The interface is updated in real-time on all activities such as messaging, appointment booking, AI queries, and accessing resources. An effective user interface boosts user experience and system efficiency (Human-Computer Interaction) (Preece et al., 2015).

Student messages are processed by the Socket.io server and saved in the MongoDB database, allowing instant delivery and notification updates in real time. The AI chatbot, which is driven by Google Gemini, will be able to give fast and precise answers, enhancing the overall user experience. Any changes made in the appointment are instantaneously conveyed throughout the platform, which makes students and staff intertwined.

The staff interface enables effective handling of appointments, communication history and learning content organization. The regular updates in real-time make the site responsive and reliable even when several users are using it at a time. Moreover, the system keeps detailed logs of all the interactions, so users can go back to the previous activities, with the correct timestamps.

4.1.3 DATA INSIGHTS

Besides easing academic interactions, the platform offers an invaluable insight into student engagement and system use. The frequency of chats, bookings done through the AI chatbot, and downloads of resources are the key performance indicators that are monitored and evaluated. These analytics will assist institutions to make an assessment of the effectiveness of the use of the academic support services available to students (Siemens & Baker, 2012).

Timestamps are recorded in all user interactions, allowing studying the patterns of engagement and the most active times. This will enable institutions to determine when academic support services are required most and to determine its effectiveness. Employees will have access to the history of their communication with students, such as chat logs, documents posted by students, and booking status.

This type of data-driven insight can assist in making better decisions and allocating resources, which will assist staff in optimizing their academic support strategies. Through such understandings, institutions are able to improve quality of the support they offer and overall student outcomes (Ferguson, 2012).

4.2 DISCUSSION

The INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM platform has been able to meet its objectives of delivering a comprehensive academic support experience that is real-time, artificial intelligence-driven doubt resolutions, scheduled slot management, and organized material distribution. The system will enable students to receive instant assistance via the AI chatbot or real time staff chat, reserve consultation sessions easily, and have access to study materials via a centralized repository. All the members of staff are able to handle all their academic support issues effectively via a special portal.

The staff and the admin dashboards are the primary control interfaces of the platform, where some significant information is presented, including active bookings, the last chat, and uploaded materials. An unread message notification in real-time, support to add files and Markdown-style AI responses are also features that improve the quality of academic interactions and minimize manual coordination. Testing results indicated that students and staff considered the platform user-friendly and effective in supporting their academic activities.

The system is also flexible and scalable, which implies that it can have more features added to it in the future without influencing the current structure. The platform can be enhanced with features like video conferencing, peer discussion forums, quiz systems, and performance analytics. This renders the system applicable in the long run in academic institutions with different sizes.

4.3 CHALLENGES AND LIMITATIONS

Although the INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM platform had worked well when tested, some challenges and limitations were pointed out. The system is dependent on the internet connection as it is based on the real-time communication through Socket.io and Google Gemini API responses, thus the system needs a consistent internet connection to be at its best. Chat delivery, as well as AI response times, can be interrupted temporarily in case of any network interruption. The

platform also relies on availability and quota restrictions of Google Gemini AI API, which can influence the results of AI chatbot during high-load times.

The second weakness is that the present version of the platform lacks support of such functions like video conferencing, push notifications, peer discussion forums and performance analytics dashboard. Such features would further improve the academic support experience, but are still to be developed. The current platform is already efficient in terms of handling a small number of parallel socket connections but would have to apply further infrastructure optimizations like load balancing and Redis-based socket clustering to support very large user bases.

Irrespective of these constraints, the platform offers a powerful and scalable base of contemporary academic support systems. Its live communication features, integration of AI, systematized slot management and organized sharing of materials make it a credible and efficient tool to use by educational institutions that want to update their academic support services.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

The INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM platform has been able to provide a complete academic support platform that integrates real-time communication, AI-based doubt solution, structured slot management, and structured sharing of study material into one unified platform. Students are able to sign in, sign up with JWT or Google OAuth, schedule a consultation with staff, chat in real-time with file sharing, communicate with an AI chatbot that is run on Google Gemini AI, and download study material. Employees will be able to create and manage consultation appointments, reply to student inquiries in real-time, upload and arrange academic materials, and track their academic support processes via a special dashboard.

The platform logs all significant interactions in real time, such as chat messages, slot bookings, AI conversations, and material uploads. The interactions are automatically recorded and mirrored in the two portals without the need to update them manually. The real-time Socket.io messages, Markdown-formatted AI replies, unread notification badges, and the ability to attach files are all features that greatly improve the quality and responsiveness of academic assistance. The use of visual cues and structured design allows the students and staff to use the platform effectively.

In testing, the system was able to process several concurrent users, proper data throughout all sessions, and provide quality real-time communication and AI responses. There were no issues with the frontend, backend and database and all the platform features worked as intended. In general, the project indicates that contemporary web technologies such as the MERN stack, Socket.io, and Google Gemini AI can be integrated to form an intelligent, responsive, and user-friendly academic support system that can greatly enhance the quality of student-faculty interaction and management of academic resources in educational establishments.

5.2 FUTURE ENHANCEMENTS

Even though the existing system can achieve its primary goals, there are a number of opportunities to enhance the platform to a greater extent. A significant improvement would be the addition of video conferencing options, which would enable students and staff to hold live online consultation sessions on-site. This would eliminate the need for external video calling tools and provide a more complete academic support experience.

One more aspect that can be improved in the platform is peer-to-peer discussion forums where students may post a question, discuss a topic with their peers, and collaboratively share knowledge. Individualized AI suggestions through the history of interactions with students may recommend the appropriate study materials and identify frequently asked questions to prepare the staff to create better academic materials. Push notifications can be implemented to notify students and staff about new messages, slot bookings, and updates to the platform.

Further enhancements can be made:

- Inclusion of an analytics dashboard that will allow administrators to track platform usage, student engagement and material popularity.
- Creation of Android and iOS native mobile apps. \
- Incorporation of automated quiz and assessment facilities to be self-assessed by students.
- New features: email and SMS-based notification support of booking confirmations and new messages.
- Introducing Redis-based Socket.io clustering to achieve better scalability with high user bases.
- Introduction of AI-based content summarization of uploaded study material.
- Participation in multiple languages to have a more inclusive academic environment. The platform can develop into an intelligent academic fully in the future.

The platform can be expanded in the future into a full-fledged intelligent academic management system that supports real-time communications and resolves doubts, as

well as offers sophisticated analytics, automated assessment, personalized learning recommendations, and full resource management. Such enhancements would enhance the system to be more intelligent, scalable and applicable in larger institutions of learning.

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APPENDIX

I. BILL OF MATERIALS

Hardware Costs:

This project does not need any additional hardware. The Student Doubt Resolution System does not require specific computers or laptops to be created and implemented; it only needs an internet-based computer or laptop. The system is web-based and does not need to utilize any special hardware apparatus.

Software Costs:

- **Frontend:** React.js, Bootstrap, JavaScript (open source)
- **Backend:** Node.js, Express.js (open source)
- **Database:** MongoDB, Mongoose (open source)
- **Real-time Communication:** Socket.io (open source)
- **Authentication:** JWT, Google OAuth, Bcrypt (open source)
- **AI Integration:** Google Gemini AI API
- **Development Tools:** Visual Studio Code, Git, GitHub, Postman (open source)

There is no software licensing on the own core technologies applied since most of the frameworks and tools are open source. The only external service is the Google Gemini API of AI, which might have charges based on the usage according to the plan chosen.

Staffing Cost:

This project does not have a cost of extra staffing. As an academic project work, the whole system was designed and developed, tested and documented separately.

II. CODE

1.Server.js (Backend Logic):

```
const express = require("express");
const cors = require("cors");
const mongoose = require("mongoose");
const dotenv = require("dotenv");

dotenv.config();
const app = express();

app.use(cors());
app.use(express.json());

mongoose.connect(process.env.MONGODB_URI)
.then(() => console.log("MongoDB Connected"))
.catch((err) => console.log(err));

app.get("/", (req, res) => {
res.send("Student Doubt Resolution System API Running");
});

app.use("/api/auth", require("./routes/authRoutes"));
app.use("/api/slots", require("./routes/slotRoutes"));
app.use("/api/chat", require("./routes/chatRoutes"));
app.use("/api/materials", require("./routes/materialRoutes"));

const PORT = process.env.PORT || 5000;
```

```
app.listen(PORT, () => {
console.log(`Server running on port ${PORT}`);
});
```

2.FrontEnd (App.jsx):

```
import React from "react";

import { BrowserRouter as Router, Routes, Route } from "react-router-dom";

import StudentLogin from "./pages/StudentLogin";

import StudentRegister from "./pages/StudentRegister";

import StudentDashboard from "./pages/StudentDashboard";

import ChatPage from "./pages/ChatPage";

import SlotBookingPage from "./pages/SlotBookingPage";

import MaterialsPage from "./pages/MaterialsPage";

import PrivateRoute from "./components/PrivateRoute";


function App() {

  return (

    <Router>

      <Routes>

        <Route path="/" element={<StudentLogin />} />

        <Route path="/register" element={<StudentRegister />} />

        <Route

          path="/dashboard"
```

```
        element={
          <PrivateRoute>
            <StudentDashboard />
          </PrivateRoute>
        }
      />
      <Route path="/chat" element={<ChatPage />} />
      <Route path="/slots" element={<SlotBookingPage />} />
      <Route path="/materials" element={<MaterialsPage />} />
    </Routes>
  </Router>
);
}

export default App;
```

III. WORK CONTRIBUTION:

Project Title: Intelligent Student Doubt Resolution System

Student Name: RAKHUL PRANAV R K
Register Number: 7376232BT175

- Created the entire front-end in React.js, with student and staff dashboards, responsive user interfaces, and navigation elements.
- Created the backend with the help of Node.js and Express to create API, manage users, and implement business logic.

- Created and deployed the MongoDB database schema to store users, slots and chat message and study materials.
- Incorporated real time chat feature with Socket.io to facilitate real time communication between students and staff.
- Introduced JWT-based secure authentication, Google OAuth, and Bcrypt-based role-based access and password security.
- Inbuilt AI chatbot assistance is based on Google Gemini AI to provide clarifications to academic doubts in real time.
- Coded the slot booking module that is used to book interactions between students and staff.
- Added the possibility of uploading and downloading study material in an organized manner to share academic resources.

As part of the process of ensuring that the systems were reliable and performed well, the following were done:

PLAGIARISM REPORT

INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM

by Rakhul Pranav R K (7376232bt175) Bt

Submission date: 17-Apr-2026 04:26PM (UTC+0530)
Submission ID: 2934829383
File name: final.pdf (913.29K)
Word count: 9726
Character count: 55028

INTELLIGENT STUDENT DOUBT RESOLUTION SYSTEM

ORIGINALITY REPORT

6%	5%	2%	6%
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49 Pages

9,726 Words

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29% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

36 AI-generated only 29%
Likely AI-generated text from a large-language model.

0 AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.